RACE Regular Analysis Through Continuous Exercise

Solid State : RACE-01

Classification of Crystalline Solids: molecular; metallic ; covalent and network solid

1. A solid is soft, good conductor of electricity and has very high melting point. Its one of the allotropic forms is the hardest known substance. Hence, the solid is an example of
(A) ionic solid (B) covalent solid (C) molecular solid (D) metallic solid
CSS191

2. The following are some statements related with solid state of matter:
(i) All crystalline solids are isotropic.
(ii) Amorphous solids are super cooled liquids with high viscosity.
(iii) In crystals, short range order exists.
(iv) Unit cell is the smallest building unit in the crystals.
(v) Every object possesses an identity element.
(vi) Each lattice point in a crystal has the same environment.
(vii) The basic unit must have the same stoichiometric composition as the entire crystal.
(viii) A crystal space lattice cannot have a positive ion at one lattice point and a negative ion at another lattice point.
The number of true statements is
CSS273

Lattice; Unit cell; Primitive Unit Cell; Seven Crystal Systems; Unit Cell shapes; Bravais lattices

3. The unit cell of lowest symmetry is
(A) cubic (B) triclinic (C) hexagonal (D) monoclinic
CSS192

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4. In a tetragonal crystal,
(A) $\alpha=\beta=90°;a=b=c$ (B) $\alpha=\beta=\gamma=90°;a=b\neq c$
(C) $\alpha=\beta=\gamma=90°;a\neq b\neq c$ (D) $\alpha=\beta=90°;=\gamma=120°;a=b\neq c$
CSS194

Packing in solids 3D; Three Dimension; Simple Cubic Closed Packing; SCC; Primitive Cubic; Density

5. FeO crystallizes in the cubic system, in which there is four formula units in each unit cell. The density of the crystal is 4.0 g/cm^3 . The side length of each unit cell is (Fe = 56, $N_A = 6 \times 10^{23}$)
(A) 4.227 Å (B) 2.424 Å (C) 4.932 Å (D) 2.974 Å
CSS202

6. The packing fraction in simple cubic lattice is
(A) $\frac{1}{6}\pi$ (B) $\frac{\sqrt{2}}{6}\pi$ (C) $\frac{\sqrt{3}}{8}\pi$ (D) $\frac{1}{2}\pi$
CSS204

7. A metallic element exists as cubic lattice. Each edge of the unit cell is 4.0 Å. The density of the metal is 6.25 g/cm^3. How many unit cells will be present in 100 g of the metal?
(A) 1.0×10^{22} (B) 2.5×10^{23} (C) 5.0×10^{23} (D) 2.0×10^{23}
CSS234

8. Aluminium crystallizes in cubic system with unit cell edge length equal to 4.0 Å. If its density is 45 $\frac{45}{16}$ cm^3, then the atomic radius of Al-atom is ($N_A = 6 \times 10^{23}$, Al = 27)
(A) 1.414 Å (B) 1.732 Å (C) 4.0 Å (D) 2.0 Å **CSS235**

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9. A solid element (monoatomic) exists in cubic crystal. If its atomic radius is 1.0 Å and the ratio of packing fraction and density is 0.1 cm^3 /g, then the atomic mass of the element is ($N_A = 6 \times 10^{23}$)
(A) 8π (B) 16π (C) 80π (D) 4π

CSS241

10. The densities of ice and water at 0°C and 1 bar are 0.96 and 0.99 g/cm^3, respectively. If the percentage of occupied space in ice is x, then the percentage of empty space in water is
(A) $\frac{32}{33}x$ (B) $\frac{33}{32}x$ (C) $100-\frac{33}{32}x$ (D) $100-\frac{32}{33}x$

CSS245

11. A metal crystallizes in such a lattice in which only 70% of the total space of the crystal is occupied by the atoms. If the atomic mass of the metal is 32π g/ mol and the atomic radius is 0.2 nm, then the density of the metal is
(A) 7.0 g/cm^3 (B) 3.5 g/cm^3 (C) 10.5 g/cm^3 (D) 14.0 g/cm^3

CSS246

Body Centred Tetragonal(BCT); Body Centred Cubic Unit Cell(BCC)

12. An element occurring in the body-centred cubic (BCC) structure has 1.208×10^{23} unit cells. The total number of atoms of the element in these cells will be
(A) 2.416×10^{23} (B) 3.618×10^{23} (C) 6.04×10^{22} (D) 1.208×10^{23}

CSS195

13. The distance between two nearest neighbours in BCC lattice of axial length ℓ, is
(A) ℓ (B) $\frac{\sqrt{3}}{2}\ell$ (C) $\frac{\sqrt{2}}{2}\ell$ (D) $\frac{1}{2}\ell$

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CSS199

14. Potassium has BCC structure with nearest neighbour distance (2.5 × 3) Å. Its density will be (K = 39, $N_A = 6 \times 10^{23}$)
(A) 1.040 kg/m^3 (B) 104 kg/m^3 (C) 520 kg/m^3 (D) 1040 kg/m^3

CSS201

15. The number of octahedral voids per unit BCC cell is
(A) 1.0 (B) 2.0 (C) 1.5 (D) 0

CSS247

16. A metal (atomic mass = 75) exist in BCC structure. If the uncovered distance between the atoms along the edge is equal to 67 pm, then the density of crystal (in g/ cm^3) is ($N_A = 6 \times 10^{23}$, $\sqrt{3} = 1.732$)

CSS277

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	5	B	B	C	A	B	A	A	C
Question	11	12	13	14	15	16				
Answer	B	A	B	D	D	2				

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Regular Analysis Through Continuous Exercise

Solid State : RACE-02

Face Centred Cubic Unit Cell(FCC); Cubic Closed Packing(CCP)

1. An alloy of copper, silver and gold is found to have copper constituting the face-centred cubic (FCC) lattice. If silver atoms occupy the edge centres and gold is present at body centre, the alloy has a formula

(A) Cu_4Ag_2Au (B) Cu_4Ag_4Au (C) Cu_4Ag_3Au (D) CuAgAu

CSS196

2. A solid PQ has rock salt type structure in which Q atoms are the corners of the unit cell. If the body-centred atoms in all the unit cells are missing, then the resulting stoichiometry will be

(A) PQ (B) PQ_2 (C) P_3Q_4 (D) P_4Q_3

CSS197

3. Xenon crystallizes in FCC lattice and the edge of the unit cell is 620 pm, then the radius of Xenon atom is

(A) 219.20 pm (B) 438.5 pm (C) 265.5 pm (D) 536.94 pm

CSS198

4. In an FCC arrangement of metallic atoms, what is the relative ratio of the ideal sizes of tetrahedral and octahedral voids?

(A) 0.543 (B) 0.732 (C) 0.414 (D) 0.637

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CSS209

5. A solid has three types of atoms, such as X, Y and Z. 'X' forms a FCC lattice with 'Y' atoms occupying all the tetrahedral voids and 'Z' atoms occupying half the octahedral voids. The simplest formula of solid is

(A) X_2Y_4Z (B) XY_2Z_4 (C) X_4Y_2Z (D) X_4YZ_2

CSS210

6. In a compound, oxide ions are arranged in CCP arrangement. Cations A occupy one-sixth of the tetrahedral voids and cations B occupy one-third of the octahedral voids. The formula of the compound is

(A) AB_2O_4 (B) ABO_3 (C) ABO_2 (D) ABO_4

CSS215

7. What is the formula of the magnetic oxide of cobalt, used in recording tapes, that crystallizes with cobalt atoms occupying one-eighth of the tetrahedral holes and one half of the octahedral holes in a closest packed array of oxide ions?

(A) Co_2O_3 (B) Co_5O_8 (C) CoO (D) Co_3O_4

CSS216

8. X-ray diffraction analysis of crystalline C_{60} (Buckminsterfullerene) shows that the crystal structure at 300 K can be regarded as FCC with a one-molecule basis and unit cell edge length of $\sqrt{2}$ nm. The molecules are orientationally disordered due to rotation. The centre-to-centre distance (in nm) between the nearest neighbour C_{60} molecules in the solid at 300 K is

CSS230

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9. Is there an expansion or contraction as iron transforms from FCC to BCC? The atomic radius of iron is 125 pm in FCC but $50\sqrt{2}$ pm in BCC.

(A) Expansion (B) Contraction

(C) Neither expansion nor contraction. (D) Unpredictable

CSS236

10. A metal exists as FCC crystal. If the atomic radius is $100\sqrt{2}$ pm and the density of metal is 12,500 kg/m^3, then the metal is (Atomic masses: Ca = 40, Co = 58.9, Sn = 119.8, Pb = 207.9 and $N_A = 6 \times 10^{23}$)

(A) Ca (B) Co (C) Sn (D) Pb

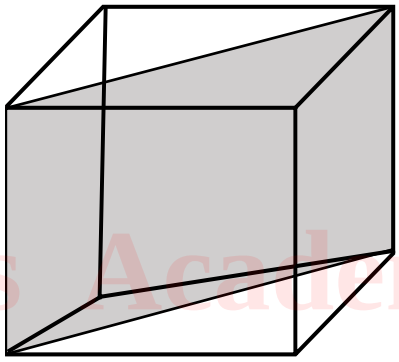
CSS240

11. In FCC unit cell, what fraction of edge is not covered by atoms?

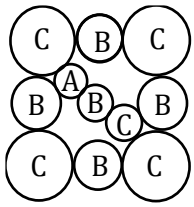
(A) 0.134 (B) 0.293 (C) 0.26 (D) 0.32

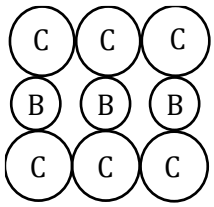
CSS242

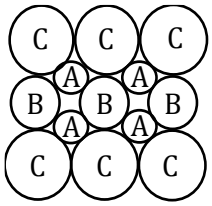
12. In a hypothetical solid, 'C' atoms form CCP lattice. 'A' atoms occupy all tetrahedral voids and 'B' atoms occupy all octahedral voids, without disturbing the crystal. If a unit cell of the crystal is cut by a plane shown in the figure, then the cross section of this plane looks like

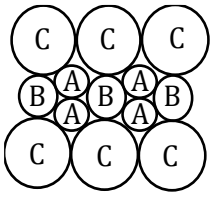


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(A)


(B)


(C)


(D)


CSS248

13. A close packing consists of a base of spheres, followed by a second layer where each sphere rests in the hollow at the junction of four spheres below it and the third layer then rests on these in an arrangement which corresponds exactly to that in the first layer. This packing is known as

(A) HCP (B) CCP

(C) square close packing (D) BCC packing

CSS251

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14. The simplest formula of a solid having CCP arrangement for 'A' atoms in which alternate face-centres are occupied by 'B' atoms and alternate edge centres are occupied by 'C' atom, is

(A) ABC (B) A_4BC (C) A_2BC (D) A_4B_2C

CSS261

15. Spinel is an important class of oxides consisting of two types of metal ions with the oxide ions arranged in CCP pattern. The normal spinel has one-eighth of the tetrahedral holes occupied by one type of metal ion and one-half of the octahedral hole occupied by another type of metal ion. Such a spinel is formed by Zn^{2+}, Al^{3+} and O^{2-}. The simplest formula of such spinel is

(A) $ZnAl_2O_4$ (B) Zn_2AlO_4 (C) $Zn_2Al_3O_4$ (D) $ZnAlO_2$

CSS262

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	A	A	A	B	D	1	B	C
Question	11	12	13	14	15					
Answer	B	C	B	B	A					

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RACE
Regular Analysis Through Continuous Exercise

Solid State : RACE-03

Voids

1. If the height of HCP unit cell of identical particles is *h*, then the height of octahedral voids from the base is

(A) $\frac{h}{2}$ (B) $\frac{h}{3},\frac{2h}{3}$ (C) $\frac{h}{4},\frac{3h}{4}$ (D) $\frac{h}{8},\frac{7h}{8}$

CSS252

2. There are three cubic unit cells A, B and C. A is FCC and all of its tetrahedral voids are also occupied. B is also FCC and all of its octahedral voids are also occupied. C is simple cubic and all of its cubic voids are also occupied. If voids in all unit cells are occupied by the spheres exactly at their limiting radius, then the order of packing efficiency would be

(A) A < B < C (B) C < A < B (C) C < B < A (D) A < C < B

CSS257

3. In a cubic closest packed structure of mixed oxides, the lattice is made up of oxide ions. Tetrahedral voids are occupied by divalent X^{2+} ions and octahedral voids are occupied by trivalent Y^{3+} ions. If only 50% of the octahedral voids are occupied, then the percentage of tetrahedral voids occupied is

(A) 50% (B) 12.5% (C) 25% (D) 40%

CSS258

4. In a compound XY_2O_4, the oxide ions are arranged in CCP arrangement and cations X are present in octahedral voids. Cations Y are equally distributed between octahedral and tetrahedral voids. The fraction of the octahedral voids occupied is

(A) 1/2 (B) 1/4 (C) 1/6 (D) 1/8

CSS264

5. The crystalline structure of a solid is CCP for 'X' atoms. If 'Y' atoms occupy all the octahedral voids and 'Z' atoms occupy all the tetrahedral voids, without any distortion in the CCP arrangement of 'X' atoms, then the fraction of body diagonal not covered by the atoms is

(A) 0.26 (B) 0.24 (C) 0.76 (D) 0.81

CSS266

6. In an ionic solid AB_2O_4, the oxide ions form CCP. 'A' and 'B' are metal ions in which one is bivalent and another is trivalent (not necessarily in given order). If all the bivalent ions occupy octahedral holes and the trivalent ions occupy tetrahedral and octahedral voids in equal numbers, then the fraction of octahedral voids unoccupied is

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{7}{8}$

CSS267

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Packing in Ionic Solids: NaCl;

7. A binary solid ($A^+ B^-$) has a rock salt structure. If the edge length is 400 pm and the radius of cation is 75 pm, then the radius of anion is

(A) 100 pm (B) 125 pm (C) 250 pm (D) 325 pm

CSS212

8. If the volume occupied by a unit cell of NaCl is 4.7×10^{-23} mL, then the volume of a crystal of NaCl weighing 1.0 g is

(A) 2.13×10^{22} mL (B) 8.03×10^{-25} mL

(C) 0.48 mL (D) 0.12 mL

CSS253

9. Solid AB has a rock salt type structure. If the radius of the cation is 200 pm, then what is the maximum possible radius of the anion?

(A) 483.1 pm (B) 273.6 pm (C) 200 pm (D) 400 pm

CSS260

10. KF crystallizes in the NaCl type structure. If the radius of K^+ ion is $\frac{186}{\sqrt{2}}$ pm and F^- ion is $\frac{214}{\sqrt{2}}$ pm, then what is the shortest distance between $K^+ - K^+$ ions (in Å)?

CSS231

ZnS; CsCl; Na_2O ; CaF_2 structure

11. An ionic solid is HCP of Q^{2-} ions and P^{x+} ions are in half of the tetrahedral voids. The value of x should be

(A) 1 (B) 2 (C) 4 (D) ½

CSS214

12. The distance between adjacent oppositely charged ions in rubidium chloride is 328.5 pm, in potassium chloride is 313.9 pm, in sodium bromide is 298.1 pm and in potassium bromide is 329.3 pm. The distance between adjacent oppositely charged ions in rubidium bromide is

(A) 314.7 pm (B) 338.5 pm (C) 339.3 pm (D) 343.9 pm

CSS259

13. The number of square faces in a truncated octahedron is 'x' and the number of carbon atoms per unit cell in diamond is 'y', then the value of (y – x) is

CSS279

Imperfection in Solid: Defects in Solids

14. The composition of a sample of cuprous oxide is found to be $Cu_{1.92}O_{1.00}$ due to metal deficient defect. The molar ratio of Cu^{2+} and Cu^+ ions in the crystal is

(A) 4 : 5 (B) 1 : 12.5 (C) 1 : 23 (D) 1 : 24

CSS219

15. The theoretical density of ZnS is d g/cm^3. If the crystal has 4% Frenkel defect, then the actual density of ZnS should be

(A) d g/cm^3 (B) 0.04d g/cm^3 (C) 0.96d g/cm^3 (D) 1.04d g/cm^3

CSS268

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16. In a crystal at 827°C one out of 10^{10} lattice site is found to be vacant but at 927°C, one out of 2×10^9 lattice sites is found to be vacant. The enthalpy of vacancy formation in kJ/mol is
(A) 76.8 (B) 176.8 (C) 33.8 (D) 23.8

CSS269

17. A metal 'M' (atomic mass = 31.25) crystallizes in CCP but it has some vacancy defect. If the edge length of the unit cell is 500 pm and the density of the metal is 1.6075 g/cm^3, then the number of moles of metal atoms missing per litre of the crystal is (1 amu = 1.67×10^{-24} g)

CSS274

18. The total number of ions per unit cell in sphalerite (zinc blend) structure if there is a Schottky defect of one anion per unit cell, is

CSS275

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	B	A	B	A	B	D	A	4
Question	11	12	13	14	15	16	17	18		
Answer	B	D	2	B	A	B	2	6		

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